

USB4 2.0 ENGINEERING CHANGE NOTICE FORM

Title: ALPM CL1 Exit Time

Applied to: USB4 Specification Version 2.0

Brief description of the functional changes:
Removed the requirement to wait CL1 Exit Time at the DP IN Adapter before enabling the DP Clock Sync process. This delay was introduced in order to avoid DP Clock Sync Packets sending while the link is in low power state, but it can't happen as the first DP Clock Sync Packet can go out after at least 2ms.

Benefits as a result of the changes:
Start DP Clock Sync process as early as possible

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:
None

An analysis of the hardware implications:
None

An analysis of the software implications:
None

An analysis of the compliance testing implications:
None

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Actual Change

(a). 10.5.7.3 Wake Sequence

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Upon detecting an LFPS, which is qualified as a wake signal according to the DisplayPort Specification, a DP IN Adapter shall send a DP Link Control Packet of type ALPM with the *S/W* bit set to 0b and the *SR Count* field set to 0h. The DP IN Adapter shall send the DP Link Control Packet within tDPALPMWake after detecting the LFPS.

When a DP IN Adapter ends its PHY establishment and detects an ML_PHY_LOCK, as defined by DisplayPort Specification, it shall start converting DisplayPort Main-Link Symbols into Tunneled Packets after receiving an SR (for a 8b/10b DP Link) or an LLCP (for a 128b/132b DP Link).

After a DP IN Adapter sends the DP Link Control Packet of type ALPM with the *S/W* bit set to 0b, it shall:

~~1. Wait for ADP_DP_CS_13. CL1 Exit Time.~~

~~2.1.~~ Before advancing to Step ~~3.2.~~, ensure it has ended its PHY establishment and detected an ML_PHY_LOCK.

~~3.2.~~ Enable its Clock Synchronization – Start advancing the Lifetime Counter and send DP Clock Sync Packets as defined in Section 10.6.

Note: Clock Synchronization is restarted from the initial state after ALPM Wake. Therefore, the first Measure Window Event will not produce a DP Clock Sync packet as it might not be accurate, from 2nd Measure Window Event and on DP Clock Sync packet shall be produced as defined in section 10.6.2.1.